

GIT 18

	:tada:
Use for the first commit in a new repository.	
:tada: start the project	
	:bug:
Correct wrong behaviour in RTL, testbench, firmware, model, or flow code.	
:bug: gemm: fix the tile count for odd partial tiles	
	:ambulance:
Repair something that blocks the whole team.	
:ambulance: widen the counters for larger workloads	
	:sparkles:
Add a new function that is complete and operational.	
:sparkles: dw: add stride-2 depthwise support	
	:hatching_chick:
Record the first time that something operates from end to end.	
:hatching_chick: dual-core hello world runs on hardware	
	:construction:
Commit work that is not complete.	
:construction: layernorm: first draft of the datapath	
	:boom:
Change an interface that other users depend on.	
:boom: csr: change the numbers of the control register fields	
	:recycle:
Change the structure of the code, but do not change its behaviour.	
:recycle: move the normalization datapath out of the output path	
	:art:
Change names, layout, or format only.	
:art: align the serializer with the stream fifo style	
	:fire:
Delete files that are no longer necessary.	
:fire: delete the old golden models	
	:coffin:
Delete code that no other code uses.	
:coffin: delete the unused normquant load state	
	:wastebasket:
Mark code, a register, or a mode for future removal.	
:wastebasket: deprecate the old identity normquant mode	
	:truck:
Move or rename files, modules, or directories.	
:truck: move the register interface into sw/	
	:rewind:
Undo a previous commit.	
:rewind: revert the pad injector change	

	:twisted_rightwards_arrows:
Merge one branch into another.	
:twisted_rightwards_arrows: merge main into devel	
	:adhesive_bandage:
Make a small correction that is not critical.	
:adhesive_bandage: add a guard around the assertion for synthesis	
	:see_no_evil:
Add or update the <code>.gitignore</code> file.	
:see_no_evil: ignore the generated waveforms	
	:bookmark:
Add a release tag, or change the version number of the repository.	
:bookmark: increment the version for the next patch release	

RTL 13

	:electric_plug:
Connect blocks that already exist.	
:electric_plug: connect clear to all engine submodules	
	:abacus:
Change how the design computes.	
:abacus: make the accumulator 40 bits wide	
	:vertical_traffic_light:
Control when data moves and which requester wins.	
:vertical_traffic_light: change the backup mode to round-robin arbitration	
	:dna:
Make a fixed value configurable.	
:dna: parameterize the input serializer for other instances	
	:control_knobs:
Change the control and status registers or the register interface.	
:control_knobs: regenerate the register interface for the new stride field	
	:bridge_at_night:
Cross a clock domain or a reset domain.	
:bridge_at_night: uart: synchronize rx_valid into the core clock	

	:shield:
Add or change protection against faults.	
:shield: interco: propagate the ecc bits through the router	
	:space_invader:
Change correct RTL because a tool cannot process it.	
:space_invader: flatten the interface array ready for elaboration	
	:fire_extinguisher:
Repair a defect that can stay in the silicon.	
:fire_extinguisher: break the combinational loop in the enable path	

	:compass:
Change how the design calculates or arranges addresses.	
:compass: addressgen: add zig-zag tile traversal	
	:satellite_antenna:
Change an off-chip link.	
:satellite_antenna: eth: handle the pause frame backoff	
	:magnet:
Work on emulation or on the laboratory setup.	
:magnet: fpga: add the board target with ddr4 and pcie	
	:label:
Add or change type definitions.	
:label: add an fsm phase type to the package	

TIMING 2

	:scissors:
Change the RTL to meet the timing target.	
:scissors: register the ready signal out of the memory	
	:triangular_ruler:
Change the constraints instead of the RTL.	
:triangular_ruler: set the io delays to 0.5*TCK and add path groups	

PPA 4

	:zap:
Decrease the number of cycles or increase the throughput.	
:zap: overlap the compute of the next tile with the current store	
	:chart_with_downwards_trend:
Decrease the number of gates, flip-flops, memories, LUTs, or FIFO entries.	
:chart_with_downwards_trend: decrease the sink fifo from 8 to 2 entries	

	:battery:
Decrease the power that the design uses.	
:battery: gate the clock of the lanes that are not active	
	:stopwatch:
Measure the performance, but do not change it.	
:stopwatch: tb: count the cycles of each load and store phase	

BACKEND 2

	:factory:
Change the implementation flow.	
:factory: add gate-level simulation to the flow	
	:cityscape:
Change the physical layout.	
:cityscape: update the floorplan for the new memory macros	

VERIFICATION 7

	:white_check_mark:
Add, extend, or repair tests and regressions.	
:white_check_mark: add partial tiling tests with stalls	

	:petri_dish:
Change the environment that runs the tests.	
:petri_dish: tb: add a timeout so that failed tests stop early	
	:game_die:
Change how you make the test data.	
:game_die: generate depthwise kernels as test specifications	
	:spider_web:
Collect or increase the coverage.	
:spider_web: exclude the testbench core from the coverage report	
	:microscope:
Make the internal behaviour visible.	
:microscope: add a transaction tracer to the interconnect	
	:safety_vest:
Add or change the checks that the design must satisfy.	
:safety_vest: assert that the bank count is a power of two	
	:rotating_light:
Correct the warnings that a tool gives.	
:rotating_light: correct the width truncation warnings	

FIRMWARE 2

	:boot:
Change the code that starts and supports the system.	
:boot: initialize the core structures at startup	
	:joystick:
Change the code that operates the hardware.	
:joystick: hal: add the stride-2 depthwise configuration	

MODELING 3

	:mirror:
Change the reference model that you compare the hardware against.	
:mirror: model the exponent packing to match the rtl	
	:monocle_face:
Examine the results.	
:monocle_face: plot the memory occupancy against the fsm state	
	:snake:
Change Python code that is not the model and not an analysis.	
:snake: divide utils.py into separate modules	

BUILD 5

	:hammer:
Change the machinery that builds and runs the project.	
:hammer: runner: rebuild the rtl only when it changes	
	:wrench:
Change the settings, not the machinery.	
:wrench: sim: increase the default testbench memory	

	:construction_worker:
Add or change the continuous integration setup.	
:construction_worker: ci: divide the build and test stages	
	:green_heart:
Repair a pipeline that fails.	
:green_heart: ci: correct the cache directory	
	:bricks:
Change the infrastructure around the flow.	
:bricks: add a cache for the regression artifacts	

DEPENDENCIES 5

	:arrow_up:
Change the version of an IP, a submodule, a tool, or a package.	
:arrow_up: bump the control ip for autotrigger support	
	:pushpin:
Change which revision you use, but not which version you want.	
:pushpin: point the dependency at the open-source release	
	:heavy_plus_sign:
Add an IP, a submodule, a package, or a tool.	
:heavy_plus_sign: add matplotlib for the analysis scripts	
	:heavy_minus_sign:
Remove an IP, a submodule, a package, or a tool.	
:heavy_minus_sign: remove the unused yaml parser	
	:alien:
Adapt to a change that you do not control.	
:alien: parse the new utilization report format	

DOCS 4

	:memo:
Add or update the documentation.	
:memo: document the supported output modes	
	:bulb:
Add or update the comments in the source code.	
:bulb: explain why this signal is registered	
	:pencil2:
Correct a spelling error in the documentation, the comments, or a message.	
:pencil2: correct the register field name in the readme	
	:page_facing_up:
Add or update the license and the file headers.	
:page_facing_up: add the license headers to the rtl sources	